Satellite Video Super-Resolution Based on Adaptively Spatiotemporal Neighbors and Nonlocal Similarity Regularization

Huan Liu, *Student Member, IEEE*, Yanfeng Gu, *Senior Member, IEEE*, Tengfei Wang, *Student Member, IEEE*, and Shengyang Li

***Abstract*— Recently, super-resolution (SR) of satellite videos has received increasing attention as it can overcome the limitation of spatial resolution in applications of satellite videos to dynamic analysis. The low quality of satellite videos presents big challenges to the development of the spatial SR techniques, e.g., accurate motion estimation and motion compensation for multiframe SR. Therefore, reasonable image priors in maximum *a posteriori* (MAP) framework, where motion information among adjacent frames is involved, are needed to regularize the solution space and generate the corresponding high-resolution frames. In this article, an effective satellite video SR framework based on locally spatiotemporal neighbors and nonlocal similarity modeling is proposed. Firstly, local prior knowledge is represented by means of adaptively exploiting spatiotemporal neighbors. In this way, implicitly local motion information can be captured without explicit motion estimation. Secondly, the nonlocal spatial similarity is integrated into the proposed SR framework to enhance texture details. Finally, the locally spatiotemporal regularization and nonlocal similarity modeling bring out a complex optimization problem, which is solved via the iterated reweighted least squares in the proposed SR framework. The videos from the Jilin-1 satellite and the OVS-1A satellite are used for evaluating the proposed method. Experimental results show that the proposed method demonstrates better SR performance in preserving edges and texture details compared with the-state-of-art video SR methods.**

***Index Terms*— Adaptively spatiotemporal neighbors, implicit motion estimation, maximum *a posteriori* (MAP), nonlocal similarity, satellite video, super-resolution (SR).**

I. Introduction

SUPER-RESOLUTION (SR) is one of the classical image restoration tasks to get the high-resolution (HR) image from one or multiple low-resolution (LR) images. Compared with the limitation of fixed size and cost of hardware, it is more popular to improve resolution by post-processing with various SR algorithms. SR has been widely used in many fields, such as HR frames reconstruction for hybrid videos and object recognition in computer vision [1], spectral restoration [2] and classification in remote sensing [3], and medical image processing [4]. According to the signal dimension of SR, it can be categorized into spatial SR, temporal SR, and spectral SR. However, SR is a severely ill-posed inverse problem since known variables of the LR image are much less than unknown variables of the HR image, where one LR image can be matched to several HR images.

Generally, SR methods can be mainly divided into two types, considering different ways to utilize variable constraints and auxiliary information.

One is based on the design of interpolation function and image degradation models. Classical interpolation methods, such as the nearest interpolation and bicubic interpolation, apply fixed interpolation patterns to images and easily cause artifacts to some extent. To overcome the limitation of interpolation, many SR methods have been developed under a unified Bayesian framework where the image degradation models are considered. Generally, the image degradation models involved in data fidelity terms are integrated with prior knowledge being regularization terms. The robustness to outliers is taken into consideration while designing data fidelity terms according to image degradation models. The basic assumption for image degradation models is that noise is the Gaussian distribution, which is measured by L_2 norm in the data fidelity term. However, it is sensitive to outliers, such as missing pixel values and motion estimation errors. Farsiu *et al*. [5] proposed to use L_1 norm to represent Laplacian noise in order to handle outliers and model uncertainty. However, it is not necessary to deal with inliers that is probable Gaussian noise. To deal with the complexity of noise distribution in real images, Lorentzian error norm [6], Huber norm [7], and half-quadratic error norm [8] are proposed to model data fidelity terms. Generally, the spatial weights in data fidelity terms are assumed to follow the uniform distribution throughout the whole image. This way does not consider the difference in different noisy regions. Kohler *et al*. [9] adaptively determined spatial-weights according to noise level, where global outliers and local outliers were respectively divided by some

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criteria. Reasonable regularization terms can help maximum *a posteriori* (MAP) converge to a reasonable solution. The regularization term is generally designed according to the properties of HR images. The Tikhonov regularization term based on the L_2 norm is one of the widely employed forms of regularizations, which opts a smooth solution by penalizing energy in the higher frequencies [10]. However, SR images using this regularization will not contain sharp edges because of high-frequency energy in edges. According to the piecewise smoothness prosperity of images, total variation (TV) [11] and bilateral-TV (BTV) [5] are respectively proposed as regularization terms to preserve sharp edges. However, this kind of regularization lacks flexibilities to characterize local structure, which also generates oversmooth results and suffers from creating undesirable artifacts in homogeneous regions. Spatial dependence between pixels can guarantee to adaptively reconstruct local edge structures. In terms of this spatial property, Markov prior [12] and nonlocal similarity prior [13] are applied for image SR. In terms of complexity in real images and videos, it is not suitable to assume that both the noise level and the blur kernel are known. Liu and Sun [14] proposed a unified Bayesian framework to simultaneously estimate the noise level, the blur kernel, and motion while reconstructing HR video frames. Babacan *et al*. [11] proposed to reconstruct HR image based on a variation Bayesian framework by jointly estimating motion parameters and unknown regularization parameters. The joint estimation is robust to deal with estimated uncertainty. Ma *et al*. [15] and Faramarzi *et al*. [7] respectively proposed to simultaneously estimate the blur kernel and generate SR videos, where the TV-L1 optical flow was used for motion estimation [16].

The other one is based on learning methods, which generally need training samples composed of a set of HR and LR image pairs. LR images in the training set are obtained by downsampling HR images to form into HR and LR image pairs. The basic assumption is that a pair of HR and LR patch shares a similar manifold structure, where how to reconstruct LR and HR image based on training sets is the same. Based on this basic assumption, Yang *et al*. [17] proposed to reconstruct a patch by sparse representation, where HR and LR dictionaries were respectively learned by training samples. To improve the representative ability of training dictionaries, training patches were clustering into several classes. And then different overcomplete dictionaries were learned from different classes, instead of one dictionary [18], [19]. In terms of the inconsistency of spare representation between LR and HR patches, Shao *et al*. [20] proposed a coupled sparse autoencoder (CSAE) to map the relation of spare representation between LR and HR patches. With large training samples, deep learning methods have been also proposed to learn the mapping relationship between LR and HR images for SR. Kim *et al*. [21] proposed a very deep convolutional neural network (CNN) for SR, termed VDSR. The LR images are initially interpolated based on the size of HR images as inputs of the network in VDSR. To avoid increasing input size of networks, an efficient subpixel convolution network (ESPCN) was proposed to accomplish upsampling by a subpixel convolution layer [22]. To enhance visual quality, Wang *et al.* [23] proposed an enhanced SR generative adversarial network (ESRGAN) by introducing residual-in-residual dense blocks. Some operations have been applied for multiframe SR, such as 3-D convolutions [24], motion compensation layers [25], and recurrent layers [26]. In terms of multiframe SR, modeling temporal dependence is one of the crucial steps, which is generally embedded into the network structure for motion estimation and motion compensation. Huang *et al*. [27] proposed a bidirectional recurrent convolutional network (BRCN) to model long-term temporal dependence by recurrent and 3-D feedforward convolutions. To generate accurate motion estimation, Wang *et al.* [28] proposed to infer HR optical flows by an optical flow reconstruction network in a coarse-to-fine manner, termed super-resolving optical flows (SOFs). Chu *et al*. [29] proposed temporally coherent generative adversarial networks (TecoGANs) for video SR by using a spatial-temporal discriminator to achieve photorealistic and temporally coherent details. Instead of using a single memory module, Wang *et al.* [30] proposed a multimemory CNN (MMCNN) by cascading an optical flow network and an image-reconstruction network. Haris *et al*. [31] proposed a recurrent backprojection network (RBPN) by treating each frame as a separate source of information to integrate spatial and temporal contexts.

With the development of remote sensing technique, several satellites can obtain remote sensing videos with a high frame rate. These videos can provide large and dynamic scenes. In the field of requiring highly dynamic changes, satellite videos provide a new means for supervision, e.g., disaster monitoring. However, the spatial resolution of satellite videos is about 1 m, which is much lower than traditional surveillance videos on the ground and UAV videos. In addition, imaging quality is severely degraded due to the impacts of noise and video compression. To enhance objects in satellite videos, it is necessary to SR for satellite videos. SR methods based on VDSR [32] and ESPCN [33] have been applied for this task. To enhance the reconstructed consistency of VDSR, Zhou *et al*. [34] added softmax loss to mean square error loss. Jiang *et al*. [35] proposed a progressively enhanced network by a transition unit to enhance residual images with fine details. To reconstruct better high-frequency components in residual images, a deep distillation recursive network (DDRN) with a multiscale purification unit was proposed for Jilin-1 video satellite image SR [36]. To recover high-frequency edges in noisy satellite videos, a GAN-based edge enhancement network (EEGAN) was proposed in [37]. However, they dealt with a single satellite video frame without consideration of temporal information.

The goal of this article is to generate HR satellite video frames from a sequence of observed frames under a Bayesian framework. Generally, satellite videos are registered among frames via platform's motion compensation for staring imaging. Therefore, in observed scenes, the main motion is caused by moving objects, such as aircrafts, ships, and vehicles. Due to large imaging scenes and low spatial resolution, these moving objects domain small and local regions compared with whole observed scenes. Therefore, global motion estimation and dense optical flow estimation is not necessary. Inspired

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by [38] and [39], spatiotemporal neighbors are used as constraints without explicit motion estimation for SR. A similarly implicit motion strategy by autoregressive (AR) model is proposed to solve this problem. Then weights of AR model are adaptively determined by a sequence of LR frames. The AR model has been successfully applied for SR, such as image and video SR [40], [41], temporal SR [42], and depth SR [43]. Recently, methods based on nonlocal similarity have been widely used in image restoration tasks, such as denoising [44], deburring [45], and SR [46]. In this article, the nonlocal similarity is also introduced as the nonlocal regularization for satellite video SR to enhance texture details.

The main contributions of the proposed method include twofolds.

1) The AR model based on adaptively spatiotemporal neighbors is exploited as a local regularization term to model implicit motion estimation. Then a sequence of LR frames adaptively determine the weights of the AR model.
2) In order to enhance texture details, a global strategy is introduced as the global regularization, which can effectively utilize the nonlocal similarity within satellite video frames.

The proposed framework is validated on videos from Jilin-1 satellite and OVS-1 satellite. The experimental results show that the proposed method achieves better SR performance than comparative methods for satellite video SR.

The remainder of this article is organized as follows. Section II describes the imaging degradation model and the proposed framework for satellite video SR. Experimental results and analysis are presented in Section III. Finally, Section IV lists conclusions and discussion of the experimental results.

II. Problem Formulation and Proposed Algorithm

In this section, the imaging degradation model is firstly described for the formulation of satellite video SR. In the proposed SR framework based on MAP, the exploitation of spatiotemporal neighbors is analyzed as a local regularization term, which implicitly estimates motion instead of explicitly estimating motion. Then the nonlocal similarity is introduced as a nonlocal regularization term to enhance texture details. Finally, the local and nonlocal regularization terms are combined for satellite videos SR.

A. Problem Formulation: Satellite Video SR

According to the imaging degradation model, there are three main degradation factors in satellite videos, including blurring, downsampling, and interfering by noise. The relationship representation between the LR and HR frame can be modeled as follows:

$$\mathbf{y} = \mathbf{SHx} + \boldsymbol{\varepsilon} \tag{1}$$

where $\mathbf{y} \in R^m$ is the observation LR frame, and $\mathbf{x} \in R^M$ is a HR frame to be reconstructed, being lexicographically ordered. m, M are respectively denoted the number pixels of the LR frame and the HR frame. $\mathbf{S}$ is a downsampling matrix, and $\mathbf{H}$ is a block-circulant blurring matrix from the point spread function (PSF) h. ε is additive noise, if Gaussian noise assumed with zero mean and β precision, then $\boldsymbol{\varepsilon} \sim N(0, \beta^{-1})$.

Assuming multiframe SR, then $\mathbf{F}$ is introduced as a wrapping matrix, representing geometrical motion between the HR frame $\mathbf{x}$ and the t-th LR frame $\mathbf{y}^{(t)}$, then the SR problem is modified as $\mathbf{y}^{(t)} = \mathbf{SHF}^{(t)}\mathbf{x} + \varepsilon^{(t)}$. For satellite videos, according to relative positions between cameras and imaging areas, there are two situations. The one is that the cameras are at rest relative to imaging area by platform's motion compensation to achieve staring imaging. The other is that the cameras are moving relative to imaging areas. The last situation can be registered in advance to be consistent with the first situation. Therefore, only the first situation is considered, where imaging cameras are at rest relative to imaging areas. In this situation, there is no global motion among video frames. In satellite videos, only local motion is considered, which is caused by moving objects, such as aircrafts, ships, and vehicles. However, owing to imaging characteristics of cameras boarded on satellites, low-contrast moving objects frequently occur in satellite videos because of moving vehicles' color, such as gray vehicles, silvery gray vehicles, and black vehicles. In addition, the noise impact is obvious. Therefore, it is difficult to estimate motion, e.g., by optical flow, and motion compensation for low-contrast moving objects, where accurate motion estimation is one of the key steps for multiframe SR. In this article, steps of explicit motion estimation and compensation are replaced with a spatiotemporal prior to model the relationship among adjacent frames, where $\mathbf{F}^{(t)}$ is assumed as an identify matrix. It is an implicit motion estimation method to deal with uncertainty caused by local motion. Therefore, the single-frame SR model $\mathbf{y} = \mathbf{SHx} + \varepsilon$ is adopted in this article. Directly solving the SR model is an ill-posed problem, as the known pixel number of $\mathbf{y}$ is less than the unknown pixel number of $\mathbf{x}$ ($m \ll M$). To deal with this problem, some constraints will be incorporated as regularization terms. Then the SR problem can be expressed as

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \|\mathbf{y} - \mathbf{SHx}\|_2^2 + \lambda \boldsymbol{\psi}(\mathbf{x}). \tag{2}$$

The first term is a data fidelity term. The second term is a regularization term. $\boldsymbol{\psi}$ reflects some prior properties of $\mathbf{x}$, e.g., sparsity of gradient profile, and λ is a regularization parameter. In the respect of Bayesian inference, the SR problem can be also interpreted by MAP estimator, which estimates the latent variable $\mathbf{x}$ given by observed variable $\mathbf{y}$ as follows:

$$\begin{aligned}\arg\max_{\mathbf{x}} p(\mathbf{x}|\mathbf{y}) &\propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \\ &= \arg\min_{\mathbf{x}} -\log p(\mathbf{y}|\mathbf{x}) - \log p(\mathbf{x})\end{aligned} \tag{3}$$

where the likelihood of the LR image $\mathbf{y}$ is equivalent to $p(\mathbf{y}|\mathbf{x}) = (\beta/2\pi)^{m/2} \exp(-\beta\|\mathbf{y} - \mathbf{SHx}\|_2^2/2)$ when assuming multivariate Gaussian noise distribution followed by $\boldsymbol{\varepsilon} \sim N(0, \beta^{-1})$. $p(\mathbf{x})$ is a prior distribution of the HR frame $\mathbf{x}$. Using the prior knowledge can improve the reconstruction performance. The two distributions in (3) are respectively corresponding to the data fidelity term and the regularization term in (2). Parameter λ in (2) controls noise level in the data fidelity term, where $\lambda = \beta^{-1}$.

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In Section II-B, reasonable regularization terms are utilized to reconstruct HR frames. On the one hand, local characterizes of HR frame are taken into consideration. Firstly, the local regularization is utilized to restore fine-detailed structures, which can preserve sharp edges. Then to avoid explicit motion estimation, implicit motion estimation among adjacent frames via the AR model will also be embedded in $\psi(\mathbf{x})$. On the other hand, remote sensing images have fined textures in homogenous regions for natural scenes. However, local regularization generally brings out oversmooth results in these regions due to unobvious edges and homogenous characteristics. Therefore, we also introduce a nonlocal regularization term to preserve fine texture details. There are many repetitive patterns throughout a remote sensing image. These repetitive patterns can enhance each other texture details. Thus, clustering similar patches gets nonlocal similarity knowledge, which formulates a nonlocal regularization term (the similar idea existed in [18] as an adaptive regularization).

B. Modeling the Locally Spatial and SpatioTemporal Regularizations

In order to model priors of $p(\mathbf{x})$, various properties of $\mathbf{x}$ have been analyzed, e.g., sparsity properties of high frequencies. Because of heavy-tailed distribution of image gradient feature, e.g., Laplacian distribution, the TV model based on image gradient feature has been successfully applied for SR. The TV model is effective in preserving edges. Similarly, the result of the Laplacian edge detector as a high-pass filter also fits the Laplacian distribution, which is more effective than the TV model to preserve sharp edges [47]. Then the local regularization can be formulated as

$$E_{\text{Edge}}(\mathbf{x}) = \sum_{j=1}^{d} \alpha_j \sum_{i=1}^{M} |\mathbf{x}_j(i)| = \sum_{j=1}^{d} \alpha_j \|\mathbf{x}_j\|_1 \tag{4}$$

where $\sum_{j=1}^{d} \alpha_j = 1$, and $\mathbf{x}_j = \mathbf{D}_j\mathbf{x}$. $\mathbf{D}_j$ is a $M \times M$ block-circulant matrix of a convolutional filter, which is determined by the edge operators, including the gradient filter and the Laplacian filter in this article. The weights of different edge operators are set equally in this article.

For multiple frames/images SR without explicit motion estimation, nonlocal means (NLM) has been extended from the image denoising problem to the image SR by splitting two subtasks in [38]. One is image deburring, where HR patches are firstly extracted before the operation of downsampling. The other one is image SR with the TV regularization. The basic assumption is that downsampling HR patches can be matched to similar LR patches in all input images. All input images are treated equally for extracting similar patches. 3-D steering kernel regression (3DSKR) has also been introduced to solve the problem by adding time difference, which is implemented in spatiotemporal neighbors when computing the steering kernel. It is different from 2-D SKR computing in spatial neighbors [39]. However, it tends to generate oversmooth results while suppressing noise. For image SR, the AR model has been proposed in [18], which is utilized as a spatially adaptive regularization term. Parameters of the AR model are determined by solving a least square problem. We extend it to a spatiotemporal regularization term to model implicit motion estimation. More specially, the object function considered in the AR model can be modeled as

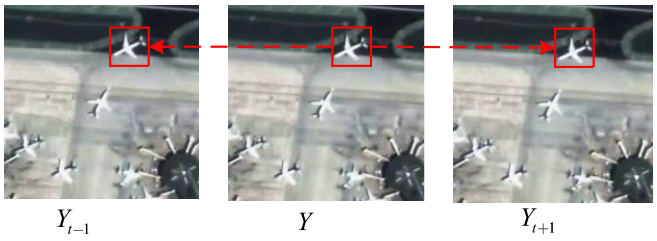


Fig. 1. Implicit motion estimation by searching best matched patches in the spatiotemporal regularization.

$$E_{\text{AR}}(\mathbf{x}) = \sum_{i} \left(x_i - \sum_{j \in \Omega(i)\backslash i} a_{i,j} x_j \right)^2 + \sum_{i} \sum_{t \in \Omega(t)} \gamma^t \left(\sum_{j \in \Omega(i)\backslash i} a_{i,j} x_j - \sum_{r \in \Omega(i)\backslash i} a^t_{i,r} x_r \right)^2 \tag{5}$$

where i, j refer to spatial locations of pixels, and $\Omega(i)\backslash i$ are spatial neighbors of the spatial location i without including itself. $\Omega(t)$ is a set of adjacent frames centered at the current frame. $a_{i,j}$ and $a^t_{i,r}$ denote AR coefficients. The first term of $E_{\text{AR}}(\mathbf{x})$ represents local similarity of HR frame $\mathbf{x}$, where a center pixel can be fit by its neighboring pixels. The coefficients $a_{i,j}$ can be computed by the current observed frame $\mathbf{y}$

$$a_{i,j} = \frac{\exp\left(-(\tilde{y}_i - \tilde{y}_j)^2/2\sigma_1^2\right)}{\sum_{j \in \Omega(i)\backslash i} \exp\left(-(\tilde{y}_i - \tilde{y}_j)^2/2\sigma_1^2\right)} \tag{6}$$

where σ_1 is a decay rate, $\tilde{\mathbf{y}}$ is from initial upsampling $\mathbf{y}$ by the bicubic interpolation, and $\tilde{y}_i, \tilde{y}_j$ are pixels of $\tilde{\mathbf{y}}$. The LR image and the HR image share similar manifold structures. Therefore, the linear fitting model based on the AR model for HR images can be assumed to share same weights with LR images.

The second term means that the reconstructed results should be similar for the same patches from different observations. It is a step to implicitly estimate motion for motion compensation. The same patches from different observations are searched by nearest neighbors within a region, which focus on current patches. By searching neighboring pixels in the adjacent frames, explicit motion estimation can be avoided when similar pixels from adjacent frames are taken into reconstructing the center pixel. The idea is similar with block matching for motion compensation. The best matched patches of a referenced patch are shown in Fig. 1, where the solid red box is the referenced patch, and the dashed red boxes are matched patches. Without explicit motion estimation, the displacements between the referenced patch and matched patches are not estimated. The coefficients $a^t_{i,r}$ are computed

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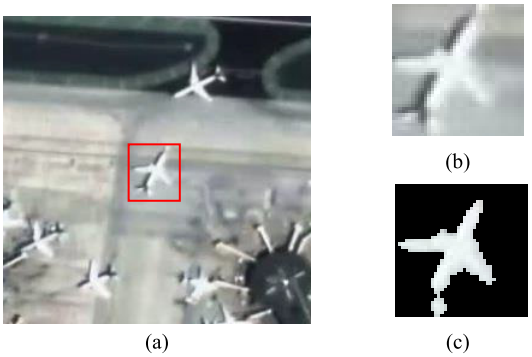


Fig. 2. Illustration of the adaptively spatiotemporal regularization. (a) Original image. (b) and (c) Enlarged parts of the red patch in (a).

by the tth LR image $\tilde{\mathbf{y}}^t$

$$a_{i,r} = \frac{\exp\left(-(\tilde{y}_i^t - \tilde{y}_r^t)^2/2\sigma_2^2\right)}{\sum_{r\in\Omega(i)\backslash i}\exp\left(-(\tilde{y}_i^t - \tilde{y}_r^t)^2/2\sigma_2^2\right)} \tag{7}$$

where σ_2 controls the decay rate of the exponential function.

The smaller the reconstructed errors, the larger the coefficients are. Therefore, the parameter γ^t should be inversely proportional to reconstructed errors between the current observation and other temporal observations

$$\gamma^t = \varepsilon \exp\left(-\left(\sum_{j\in\Omega(i)\backslash i} a_{i,j}x_j - \sum_{r\in\Omega(i)\backslash i} a_{i,r}^t x_r\right)^2 \Big/ 2\sigma_3^2\right) \tag{8}$$

where σ_3 adjusts the importance of reconstructed difference, and ε is the weight decay.

The fitting errors of AR can be assumed as Gaussian distribution. For notional simplicity, a matrix form of $E_{\mathrm{AR}}(\mathbf{x})$ is formulated as follows:

$$E_{\mathrm{AR}}(\mathbf{x}) = \|(\mathbf{I}-\mathbf{A})\mathbf{x}\|_2^2 + \sum_t \gamma^t \|(\mathbf{A}-\mathbf{A}^t)\mathbf{x}\|_2^2 \tag{9}$$

where $\mathbf{I}$ is an identify matrix and

$$\mathbf{A}(i,j) = \begin{cases} a_{i,j}, & \text{if } x_j \text{ is an element of } \mathbf{x}_{\Omega(i)\backslash i} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Generally, spatial neighbors of pixel x_i are represented by a fixed patch $\Omega(i)\backslash i$, which unavoidably introduce nonhomogeneous pixels. An example is shown in Fig. 2(b) for illustration. The center pixel belongs to an airplane, which is much different compared with the background of soil. The fixed patch of neighbors introduces large reconstructed errors in this situation. In this article, adaptive neighbors are introduced, which are determined by superpixel segmentation. In a fixed patch, if both a pixel and the center pixel x_i belong to the same superpixel, then the pixel is reserved as a member of neighbors. Otherwise, the pixel is discarded. The discarded pixels are the black pixels in Fig. 2(c), and the other pixels are reserved as spatial neighbors $\Omega(i)\backslash i$.

C. Modeling the Nonlocal Regularization

In addition to locally spatial regularization terms, nonlocal similarity also contributes to SR problem. It is based on the fact that similar texture regions of a small fixed region are repetitively found in the other image regions. According to this property, example-based synthesis [48], nonlocal similarity-based regularization [18], and low-rank based regularization [49] have been developed for the SR task. The local regularization can preserve shape edges, while it also probably oversmooths fine texture structures in the internal regions among edges. Nonlocal similarity can leverage the fine structure captured by a relatively small number of pixels to keep fine textures. Inspired by this idea, a nonlocal regularization is adopted as follows:

$$E_{\mathrm{NL}}(\mathbf{x}) = \sum_{\mathbf{x}_n\in\mathbf{x}} \left(\left\| \mathbf{x}_n - \sum_{l=1}^{L} r_n^l \mathbf{x}_n^l \right\|_2^2 \right) \tag{11}$$

where $\mathbf{x}_n$ is a local patch centered at pixel x_n in $\mathbf{x}$, $\{\mathbf{x}_n^1, \ldots \mathbf{x}_n^l, \ldots \mathbf{x}_n^L\}$ are the similar patches of $\mathbf{x}_n$ by searching L similar patches in the whole image $\mathbf{x}$. The measure of similar patches can be computed by the Euclidean distance $d_n^l = \|\mathbf{x}_n - \mathbf{x}_n^l\|_2^2$. The weight r_n^l can be computed by

$$r_n^l = \frac{\exp\left(-d_n^l/2\sigma^2\right)}{\sum_{l=1}^{L}\exp\left(-d_n^l/2\sigma^2\right)} \tag{12}$$

where σ is a decaying factor of the weight. For convenience of expression, let $\mathbf{R}_n$ denote the extracting matrix, where the extracted patch from image $\mathbf{x}$ can be represented as $\mathbf{x}_n = \mathbf{R}_n\mathbf{x}$

$$\begin{aligned} E_{\mathrm{NL}}(\mathbf{x}) &= \sum_{\mathbf{x}_n\in\mathbf{x}} \left(\left\| \mathbf{R}_n\mathbf{x} - \sum_{l=1}^{L} r_n^l \mathbf{R}_n^l \mathbf{x} \right\|_2^2 \right) \\ &= \sum_{\mathbf{x}_n\in\mathbf{x}} \|\mathbf{G}_n\mathbf{x}\|_2^2 \end{aligned} \tag{13}$$

where $\mathbf{G}_n = \mathbf{R}_n - \sum_{l=1}^{L} r_n^l \mathbf{R}_n^l$.

Putting all local and nonlocal regularization terms together, then the satellite video SR problem can be formulated as

$$\begin{aligned} \mathbf{x} = \arg\min_{\mathbf{x}} &\|\mathbf{y} - \mathbf{S}\mathbf{H}\mathbf{x}\|_2^2 + \lambda \sum_{j=1}^{d} \alpha_j \|\mathbf{D}_j\mathbf{x}\|_1 \\ &+ \eta\|(\mathbf{I}-\mathbf{A})\mathbf{x}\|_2^2 + \eta \sum_t \gamma^t \|(\mathbf{A}-\mathbf{A}^t)\mathbf{x}\|_2^2 \\ &+ \mu \sum_{\mathbf{x}_n\in\mathbf{x}} \|\mathbf{G}_n\mathbf{x}\|_2^2. \end{aligned} \tag{14}$$

To use gradient-based methods, the L_1 norm is approximated with a differentiable function $\varphi(x^2) = \sqrt{x^2+\tau}(\tau = 10^{-4})$. Then all the terms in the objective function are L_2-based.

Equation (14) can be solved by iterated reweighted least squares (IRLSs), which iteratively solves the following linear

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(a) (b)

Fig. 3. Sample frames of satellite videos. (a) Sample frames in the Jilin-1 data set, including Airplane, Highway, Intersection, and Overpass. (b) Sample frames in the OVS-1A data set, including Scene1 and Scene2.

Algorithm 1 Satellite Video Super-Resolution Based on the Proposed Framework

Input: the low resolution satellite video frame $\mathbf{y}$, its adjacent frames $\{\mathbf{y}_{-t}, \ldots, \mathbf{y}_t\} \backslash \mathbf{y}$ and upscaling factor s, Parameters λ, η, μ, Maximum iteration T, Blur matrix $\mathbf{H}$, and Stopping criteria $\delta = 0.001$. Output: HR frame $\mathbf{x}$
Initialization: obtain initial estimation $\mathbf{x}^{(0)}$ via bicubic interpolation.
while $\left\|\mathbf{x}^{(t-1)} - \mathbf{x}^t\right\| \geq \delta$ and $k \leq T$ do
Update $\mathbf{x}^t$ using Eq. (15).
$k \leftarrow k + 1$
end while

system:

$$\left[\mathbf{H}^T\mathbf{S}^T\mathbf{S}\mathbf{H} + \lambda\sum_{j=1}^{d}\alpha_j\mathbf{D}_j^T\mathbf{W}_j\mathbf{D}_j + \eta(\mathbf{I}-\mathbf{A})^T(\mathbf{I}-\mathbf{A}) + \eta\sum_t r^t(\mathbf{A}-\mathbf{A}^t)^T(\mathbf{A}-\mathbf{A}^t) + \mu\sum_{\mathbf{x}_n\in\mathbf{x}}\mathbf{G}_n^T\mathbf{G}_n\right]\mathbf{x} = \mathbf{H}^T\mathbf{S}^T\mathbf{y} \quad (15)$$

where $\mathbf{W}_j$ is a diagonal matrix, and every element $w_{i,i}$ is equal to $1/((\mathbf{D}_j\mathbf{x})_i^2 + \tau)^{1/2}$. Finally, we can get an optimal solution for satellite video SR by Algorithm 1.

III. Experimental Setup and Results

The proposed method is tested on videos from Jilin-1 satellite and OVS-1A satellite. Description of the data sets and experimental settings are given in Sections III-A–III-E. Then the SR results are discussed.

A. Data Preparation

The first satellite video data set is from Chinese Jilin-1 series, which provide about 1-m spatial resolution videos with 25 frames/s. The duration is focused on 30∼120 s. In terms of large scenes and primarily interested dynamic objects, only parts of video blocks are extracted from original satellite videos to conventionally test the proposed method. All the frame size in video blocks is set as $256 \times 256 \times 3$, and the number of frames is set as 64 frames. Total of four scenes are selected for Jilin-1 data set, including Airplane, Highway, Intersection, and Overpass. The sample frames of satellite videos are shown in Fig. 3(a). Consistent with most methods, RGB color space is converted into YCbCr color space. Only the Y luminance channel in YCbCr color space is used for experiments to test SR methods, since humans are more sensitive to illumination changes. And the color channels (Cb, Cr) are upscaled by bicubic interpolation. The peak signal-to-noise ratio (PSNR) and the structural similarity (SSIM) [50] of SR frames on Y luminance channel are used for evaluating performance.

The second video data set is from OVS-1A satellite,[1] which is directly used for testing on real satellite videos instead of downsampling HR frames as inputs, which is done in simulated experiments on the Jilin-1 data set. The spatial resolution is 1.98 m, and the frame rate is 20 frames/s. Two video blocks are used to test the proposed method as shown in Fig. 3(b) respectively termed Scene1 and Scene2. All the frame size in these two video blocks is $148 \times 148 \times 3$.

B. Experimental Setup

In the simulated experiments on the Jilin-1 data set, the LR frames were generated by firstly applying a 15×15 Gaussian blur kernel with two standard deviation and then downsampling by a scaling factor of 4. As for changeable SR tasks, we only compared the SR results for the upscaling factor of 4.

We compare the performance of the proposed method with state-of-art multiple images of SR algorithms, including 3DSKR [39], Bayesian-Mb [15], and TecoGAN [29]. The public implementations download from the authors' websites are used. 3DSKR [39] exploits spatiotemporal neighbors by 3DSKR for video SR, which is a similar idea with the proposed method. Bayesian-Mb [15] is one of the classical video SR algorithms by simultaneously reconstructing SR frames and handling kernel blur estimation, which is based on explicit motion estimation and motion compensation by TV-L1 optical flow [16]. TecoGAN is based on adversarial training for video SR, where a flow estimation network is used for motion compensation. In addition, the Bicubic interpolation method is also used as a baseline method. The size of spatiotemporal neighbors in 3DSKR has adopted the default set as $7 \times 7 \times 5$, where the temporal length is 5, and the spatial size is 7×7. Since outputs of 3DSKR have invalid pixels near the frame boundaries, we computed PSNR and SSIM only in the center region with shaved frame boundaries for all SR methods. And the shaved length is 22 for each boundary for upscaling factor of 4. The length of temporal

[1] https://www.myorbita.net/wxsp/list_52.aspx

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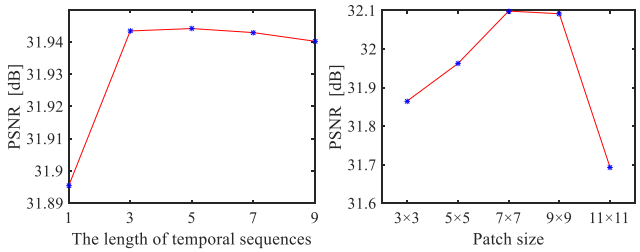


Fig. 4. Influence of parameters for the 21st frame in the "Airplane" scene video from the Jilin-1 data set. (a) Influence of the length of temporal sequences. (b) Influence of patch size.

sequence is also set as default 31 in Bayesian-Mb. A temporal length of 10 is used in TecoGAN. In the proposed method, ten nearest patches are searched for each $\mathbf{R}_n\mathbf{x}$. Simple linear iterative clustering (SLIC) [51] is used for the superpixel segmentation to adaptively determine spatial neighbors.

C. Parameter Analysis

In the proposed framework, five free parameters need to be tuned, including regularization parameters λ, η, μ in (15), the length of temporal sequences in the spatiotemporal regularization term, and the patch size of $\mathbf{R}_n\mathbf{x}$ in the nonlocal regularization term.

The multiparameter discrepancy principle is firstly used to choose regularization parameters λ, η, μ, which uses an approximate model function to iteratively choose local optimal values of regularization parameters λ, η, μ. More details can be found in [52, Sec. 4]. In consideration of experimental complexity, regularization parameters are only iteratively chosen in the 21st frame in the "Airplane" scene video from the Jilin-1 data set. The regularization parameters are $\lambda = 6.23 \times 10^{-6}, \eta = 8.24 \times 10^{-4}, \mu = 5.72 \times 10^{-6}$ when convergence. These regularization parameters are fixed in the following experiments.

To determine the length of temporal sequences in the spatiotemporal regularization term and the patch size in the nonlocal regularization term, the 21st frame in the "Airplane" scene video from the Jilin-1 data set is used for simulation. To determine the length of temporal sequences in the spatiotemporal regularization term $E_{\mathrm{AR}}(\mathbf{x})$, only the locally spatial regularization term $E_{\mathrm{Edge}}(\mathbf{x})$ and $E_{\mathrm{AR}}(\mathbf{x})$ are used for experiments. The SR frame is recovered with various lengths of temporal sequences, e.g., 1, 3, …,9. As shown in Fig. 4(a), the PSNR values do not change much when the length of temporal sequences is longer than 3. Therefore, we set the length of temporal sequence as 3 in the spatiotemporal regularization term (two adjacent frames, one is the forward frame, and the other one is the backward frame) in the following experiments. To determine the patch size in the nonlocal regularization term $E_{\mathrm{NL}}(\mathbf{x})$, only the locally spatial regularization term $E_{\mathrm{Edge}}(\mathbf{x})$ and $E_{\mathrm{NL}}(\mathbf{x})$ are used for experiments. The SR frame is recovered with various patch sizes, e.g., $3 \times 3, 5 \times 5, \ldots, 11 \times 11$. As shown in Fig. 4(b), it has highest PSNR value with the 7×7 patch size. When the patch size is smaller than 7×7, the PSNR value is increasing with the patch size, while larger the patch size will let down the SR performance. In the following experiments, the patch size is set as 7×7 in the nonlocal regularization term.

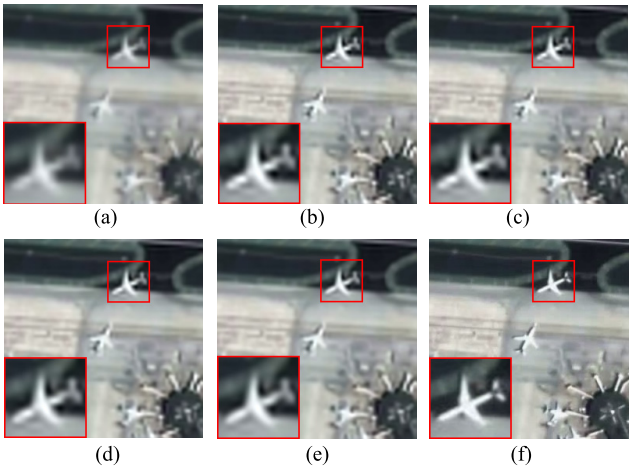


Fig. 5. Comparison of SR results (upscaling factor of 4) on the 26th frame in the "Airplane" scene by the proposed method. The red boxes are zoomed for visualization. (a) Bicubic (PSNR = 28.7779 and SSIM = 0.8801). (b) Edge (PSNR = 31.7297 and SSIM = 0.9286). (c) Edge-AR (PSNR = 31.8667 and SSIM = 0.9306). (d) Edge-NL (PSNR = 31.8312 and SSIM = 0.9298). (e) Edge-AR-NL (PSNR = 32.2419 and SSIM = 0.9345). (f) Original frame.

D. Experimental Results on the Jilin-1 Data Set

To verify the effectiveness of the spatiotemporal regularization term and the nonlocal regularization term proposed in our framework, the SR results on the 26th frame in the Airplane satellite video are first presented in Fig. 5. The SR frames are reconstructed only using the locally spatial regularization term $E_{\mathrm{Edge}}(\mathbf{x})$, termed Edge. Only $E_{\mathrm{Edge}}(\mathbf{x})$ and the spatiotemporal regularization term $E_{\mathrm{AR}}(\mathbf{x})$ are used for SR, termed Edge-AR. And only $E_{\mathrm{Edge}}(\mathbf{x})$ and the nonlocal regularization term $E_{\mathrm{NL}}(\mathbf{x})$ are used, termed Edge-NL. All the regularization terms are combined in our proposed framework, termed Edge-AR-NL. More PSNR and SSIM results can be found in Table I, where PSNR and SSIM values are averaged over six frames for various algorithms at each testing video. The best results are shown in bold. From Fig. 5 and Table I, we can see that the spatiotemporal regularization is effective for satellite video SR. By combing with the nonlocal regularization, the SR performance can be further improved to enhance texture details in homogenous regions.

Compared with SR methods based on 3DSKR [39], Bayesian-Mb [15], and TecoGAN [29], the proposed method shows higher PSNR and SSIM in most cases for the Jilin-1 data set as shown in Table I. A state-of-the-art deconvolution method [53] is used to postprocess 3DSKR outputs of the original implementation, which does not include the deblurring step. The 3DSKR method accomplishes implicit motion estimation by 3-D steering kernel regression, which shows little low PSNR and SSIM values. It is because it tends to oversmooth while suppressing noise. And fixed neighborhood patches introduce large reconstructed errors while treating edges for satellite video frames. With the known Gaussian

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TABLE I

AVERAGE PSNR (dB) AND SSIM OF DIFFERENT METHODS ON THE JILIN-1 DATA SET (UPSCALING FACTOR OF 4)

PSNR	Bicubic	3DSKR	Bayesian-Mb	TecoGAN	Edge	Edge-AR	Edge-NL	Edge-AR-NL
Airplane	28.8042	30.6845	31.9811	32.1331	31.8591	31.9742	31.9444	**32.3485**
Highway	30.2679	32.3499	30.8378	33.5483	33.4473	33.4809	33.5272	**33.6277**
Intersection	26.9027	29.0336	28.8788	29.7312	30.6909	30.7124	30.7662	**30.9446**
Overpass	32.4790	34.9071	33.2919	36.2619	36.6227	36.6474	36.8249	**37.0148**
Average	29.6135	31.7438	31.2474	32.9186	33.1550	33.2037	33.2657	**33.4839**
SSIM	Bicubic	3DSKR	Bayesian-Mb	TecoGAN	Edge	Edge+AR	Edge+NL	Edge+AR+NL
Airplane	0.8817	0.9125	0.9306	0.9331	0.9306	0.9323	0.9317	**0.9362**
Highway	0.8510	0.8860	0.8722	**0.9164**	0.9050	0.9057	0.9070	0.9081
Intersection	0.7921	0.8440	0.8604	0.8747	0.8872	0.8877	0.8912	**0.8958**
Overpass	0.8957	0.8604	0.8747	**0.9505**	0.9454	0.9458	0.9476	0.9502
Average	0.8551	0.8758	0.8845	0.9187	0.9171	0.9179	0.9194	**0.9226**



(a) (b) (c) (d) (e) (f)

Fig. 6. Visual results using different reconstructed methods on the 22nd frame on the "Airplane" scene for upscaling factor of 4. The red boxes are zoomed for better visualization. (a) Bicubic (PSNR = 28.8557 and SSIM = 0.8834). (b) 3DSKR (PSNR = 30.7564 and SSIM = 0.9142). (c) Bayesian-Mb (PSNR = 32.1372 and SSIM = 0.9330). (d) TecoGAN (PSNR = 32.1795 and SSIM = 0.9340). (e) Proposed method (PSNR = 32.4466 and SSIM = 0.9375). (f) Original frame.



(a) (b) (c) (d) (e) (f)

Fig. 7. Visual results using different reconstructed methods on the 22nd frame on the "Highway" scene for upscaling factor of 4. The red boxes are zoomed for better visualization. (a) Bicubic (PSNR = 30.2972 and SSIM = 0.8501). (b) 3DSKR (PSNR = 32.2915 and SSIM = 0.8848). (c) Bayesian-Mb (PSNR = 30.8211 and SSIM = 0.8714). (d) TecoGAN (PSNR = 33.6083 and SSIM = 0.9154). (e) Proposed method (PSNR = 33.4653 and SSIM = 0.9073). (f) Original frame.

blur kernel with two standard deviation in the simulated experiments, the Bayesian-Mb method estimates the HR frame with explicit motion estimation using the dense optical flow. For the "Airplane" scene, it has a high PSNR value, where estimated errors are small. However, it has large estimated errors for low-contrast moving objects. Therefore, the Bayesian-Mb has lower PSNR and SSIM values than other comparative methods for the "Highway," "Intersection," and "Overpass" scenes, where small moving vehicles decrease the accuracy of motion compensation with bias in motion estimation.

The TecoGAN method is based on the deep learning framework, which shows the great ability for satellite video SR. In "Highway" and "Overpass" scenes on the Jilin-1 data set, the proposed method has little lower SSIM values than the TecoGAN method, but higher PSNR values than the TecoGAN method. To visualize the performance for an upscaling factor of 4 SR, the visualization results for the 22nd frame in video blocks are shown in Figs. 6–9. Invalid pixels in the outputs of 3DSKR are filled with black pixels. As we can see, the proposed method shows better visual performance and better ability to preserve sharp edges.

For large objects, e.g., airplane, and repetitive texture patterns, e.g., road, the proposed method shows higher contrast compared with other methods, which are signed by red boxes in Figs. 6(e) and 9(e). It is consistent with our proposed local and nonlocal regularization terms. The neighboring information and nonlocal similar textures provide useful SR information.

For the "Overpass" scene of video block, our proposed method also has sharp edges, where artifacts are also intro-

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Fig. 8. Visual results using different reconstructed methods on the 22nd frame on the "Intersection" scene for upscaling factor of 4. The red boxes are zoomed for better visualization. (a) Bicubic (PSNR = 26.9339 and SSIM = 0.7946). (b) 3DSKR (PSNR = 29.1064 and SSIM = 0.8452). (c) Bayesian-Mb (PSNR = 29.0236 and SSIM = 0.8627). (d) TecoGAN (PSNR = 29.7674 and SSIM = 0.8764). (e) Proposed method (PSNR = 30.9801 and SSIM = 0.8969). (f) Original frame.

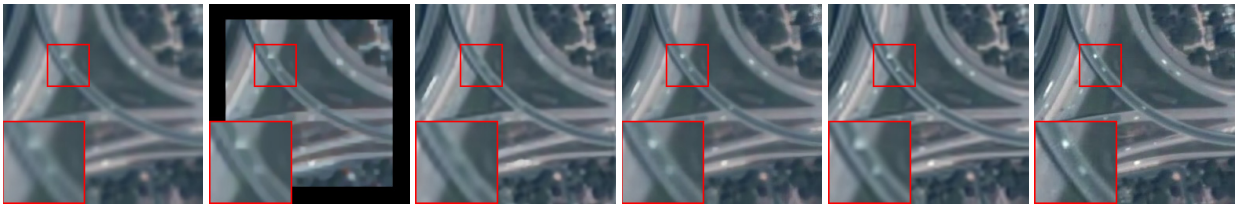


Fig. 9. Visual results using different reconstructed methods on the 22nd frame on the "Overpass" scene for upscaling factor of 4. The red boxes are zoomed for better visualization. (a) Bicubic (PSNR = 32.3947 and SSIM = 0.8934). (b) 3DSKR (PSNR = 34.7852 and SSIM = 0.9252). (c) Bayesian-Mb (PSNR = 32.1242 and SSIM = 0.8885). (d) TecoGAN (PSNR = 36.1680 and SSIM = 0.9489). (e) Proposed method (PSNR = 36.9370 and SSIM = 0.9484). (f) Original frame.

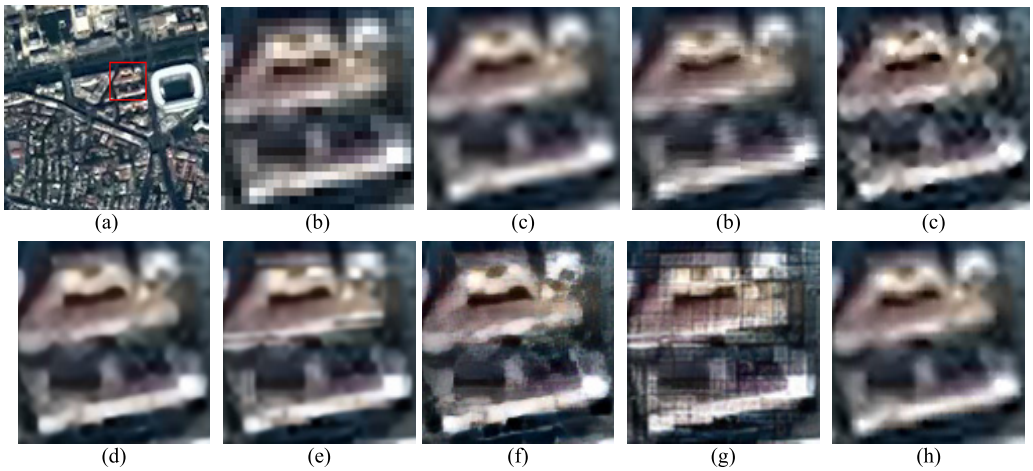


Fig. 10. Visual results using different reconstructed methods on the 21st frame on the Scene1 video for the upscaling factor of 4. The red boxes are zoomed for better visualization. (a) LR input frame. (b) Nearest. (c) Bicubic. (b) 3DSKR. (c) Bayesian-Mb. (d) SOF. (e) RBPN. (f) TecoGAN. (g) ESRGAN. (h) Proposed.

duced. It is because the noise distribution is different among adjacent frames and regions of the center frame, which will cause regression errors. As the SR results of Bicubic interpolation shown in Figs. 7(a), 8(a), and 9(a), the moving vehicles are almost been blurred, which have very low contrast with the background. It brings huge challenges for SR especially for methods by explicit motion estimation. The Bayesian-Mb method shows poor SR performance in the "Highway," "Intersection," and "Overpass" scenes as shown in Figs. 7(c), 8(c), and 9(c), where moving vehicles are wrongly reconstructed. The TeoGAN method shows sharper edges for manufactured objects, but also generates false textures as shown in Fig. 7(d), and oversmooths the SR results as shown in Figs. 8(d) and 9(d). Due to the serious loss of object information and without any reasonable referenced objects, all the SR methods show the limitation for small moving objects.

E. Experimental Results on the OVS-1A Data Set

Besides the Bicubic interpolation, 3DSKR [39], Bayesian-Mb [15], and TecoGAN [29], to test the performance of the proposed method in the real videos from the OVS-1A satellite, additional three deep learning methods for video SR are also added for comparison, including SOF [28], RBPN [31], and ESRGAN [23]. SOF [28] inferred HR optical flows in a coarse-to-fine manner instead of estimating LR optical flows by input frames for video SR. The processes of SR and

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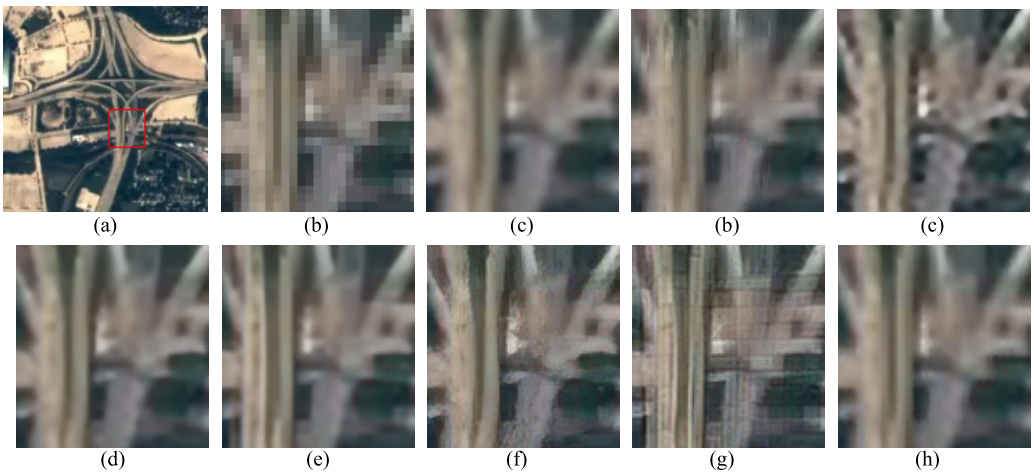


Fig. 11. Visual results using different reconstructed methods on the 21st frame on the Scene2 video for the upscaling factor of 4. The red boxes are zoomed for better visualization. (a) LR input frame. (b) Nearest. (c) Bicubic. (b) 3DSKR. (c) Bayesian-Mb. (d) SOF. (e) RBPN. (f) TecoGAN. (g) ESRGAN. (h) Proposed.

estimating HR optical flows were embedded into an end-to-end deep learning network. RBPN [31] fused multiframe information by the recurrent neural network and explicit motion estimation, which iteratively refined the network. ESRGAN is a single frame SR method, which uses an improved generative adversarial network for SR. The public implementations and trained models are downloaded from the authors' websites for comparison. In these two real videos from the OVS-1A satellite, all degraded factors are unknown. In the proposed framework, the blur kernel matrix is firstly estimated by [53], where the dark channel prior is used for deblurring. With the unknown blur kernel matrix, the Bayesian-Mb method simultaneously estimates the blur kernel matrix and reconstructs the HR frame.

The visual performance of different methods for satellite video SR is shown in Figs. 10 and 11. The SR results of 3DSKR, Bayesian-Mb, TeoGAN, and ESRGAN generate lots of obvious artifacts. ESRGAN is a single frame SR method, which has worse visual performance than multiframe SR methods. According to query results of Google Maps, RBPN generates obviously nonexistent textures, e.g., the widely left road in the zoom region in the Scene2 video as shown in Fig. 11(e). In this region, the results of SOF and the proposed method are more approximated to the true situation as shown in Fig. 11(d) and (h). SOF oversmooths the SR results when suppressing noise as shown in Fig. 10(d). In both two videos, the proposed method has better visual performance as shown in Figs. 10 and 11.

IV. Conclusion

In this article, a new approach for satellite video SR is presented. The proposed method is based on MAP by introducing two local and nonlocal regularizations. The local regularization based on the AR model implicitly estimates motion, where different temporal observations for the same region have similar reconstruction results. And the nonlocal regularization is based on nonlocal similarity to enhance texture details. The experimental results on the Jilin-1 data set and OVS-1A data set show that the proposed regularizations are efficient by using local and nonlocal information for SR, especially for large objects and repetitive texture patterns. The proposed method shows higher PNSR values and better visual performance than the comparative video SR methods for satellite videos. Moreover, the proposed method and comparative methods have limited ability for small objects, which domain several pixels. The comparative deep learning methods are also limited for SR of small moving objects, which did not especially consider this situation during training phases. However, how to make full use of the local and nonlocal region information with auxiliary knowledge is still the key step for satellite video SR. The learning-based methods use extra auxiliary information by training HR-LR image pairs, which are a possible solution for small and single objects SR. In addition, in terms of the inaccurate estimation of PSF in actual applications, more attention will be paid to some learning-based SR algorithms in the future, which are blind with respect to PSF.

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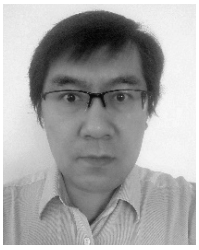
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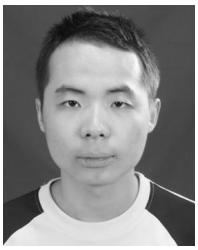
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